Report POC Deployment Image WebSocket to CCE Huawei Cloud (Region Singapore)

Report Date : 30-04-2025
Project Name : Deployment Image WebSocket to CCE Huawei Cloud
Project Detail : Deployment Image WebSocket to CCE Huawei Cloud

Resource

Node Pool

No	Name	Region	AZ	Flavor	vCPUs	Memory (GiB)	OS	Bandwith Type
1	Nodepool-messaging	Singapore	AZ1 -AZ5	C7n.2xlarge.2	8	16	Ubuntu 20.4	300Mbit/s | Exclusive | Shared
2	Nodepool-base	Singapore	AZ1- AZ5	C7n.2xlarge.2	8	16	Windows Server 2016 Datacenter 64bit	300Mbit/s | Exclusive | Shared

Cluster

Kategori	Parameter	Nilai
Basic Settings	Type	CCE Turbo
	Billing Mode	Yearly/Monthly | 1 month | Auto-renewal
	Cluster Name	Mandatory
	Cluster Version	v1.30
	Cluster Scale	Nodes: 50
	Master Nodes	3 Masters
	Master Distribution	Automatic
Network Settings	Network Model	Cloud Native Network 2.0
	VPC	Mandatory

	IPv6	Close
	Default Node Security Group	Auto generate
	DataPlane V2	Close
	Default Security Group	Auto generate
	Service CIDR Block	10.247.0.0/16
	Request Forwarding	iptables
Advanced Settings	Overload Control	Enable
	Time Zone	(UTC+08:00) Asia/Shanghai
Add-ons	CCE Container Network	Default Installation

Ingress/LoadBalancer

Bagian	Field	Nilai / Pilihan
General	Name	ingress-messaging
	Namespace	gandalfus-messaging
Load Balancer	Load Balancer Type	Shared
	Auto Create	✓ (enabled)
	Instance Name	elb-messaging-kubernetes
	Guaranteed Performance	✕ (not selected)
	EIP	Auto create
	Line	Dynamic BGP
	Billing Mode	Traffic (for light/bursting traffic)
	Bandwidth Options	1, 5, 10, 50, 100, 300 Mbps (custom selection area)

	Selected Bandwidth	300 Mbps
	Resource Tag	- Tag Management Services (TMS)- View TMS Tag List- Add tag (optional)
Listener	Frontend Protocol	HTTP (selected), HTTPS (not selected)
	External Port	Auto create (with note: route maintains external port config of the created listener)
	Access Control	Inherit ELB Configurations
	Advanced Options	Add advanced options (optional)
Footer	Notes	I have read Notes on Using Load Balancers
	Button	OK

Work Detail

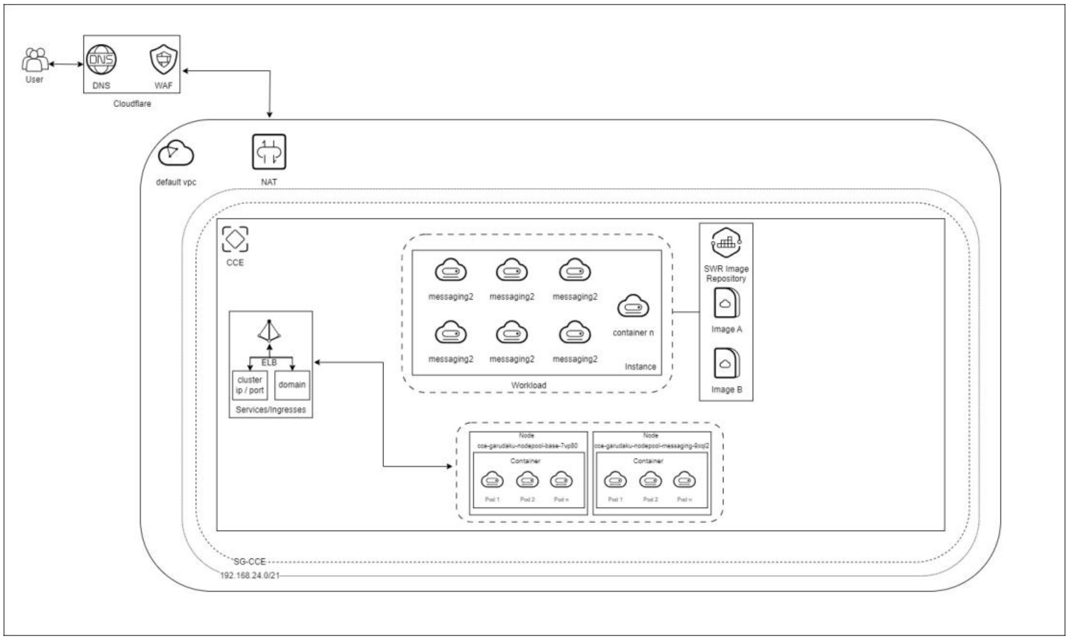
Planned start datetime	25-02-2025 08.30
Planned finish datetime	14-03-2025 23:00
Actual start datetime	25-02-2025 08:30
Actual finish datetime	14-03-2025 23:59
Delay Reason	.
Resource Used	CCE, ELB, EIP, ECS, SWR
Client Confirmation Time	14-03-2025
Release Time	

Client Detail (PIC)

Name	Phone	Organization
Mr Subangkit	+62 811-1353-979	Garudaku

Dokumentasi

Topologi



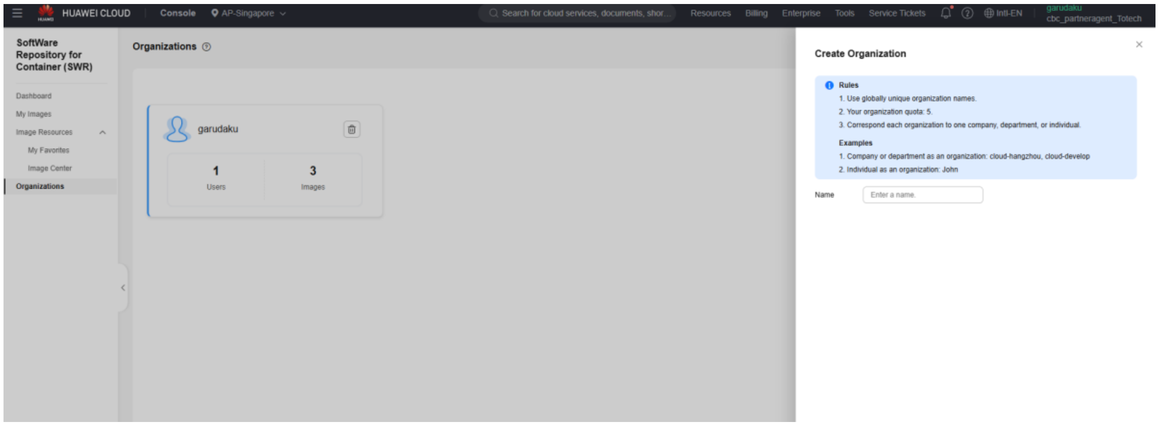
1. Prasyarat

- Akun Huawei Cloud aktif
- Image WebSocket di lokal (Docker)
- Akses ke Huawei Cloud SWR dan CCE
- Huawei Cloud CLI atau akses ke Console

2. Push Image WebSocket ke SWR Huawei Cloud

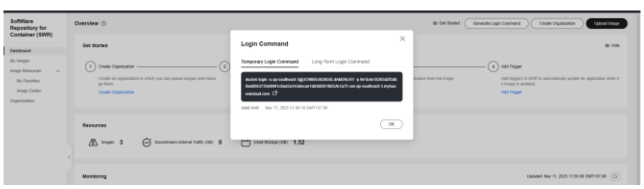
Langkah 1: Login ke SWR Huawei Cloud dan Buat Organization

1. Login ke Huawei Cloud Console melalui https://console.huaweicloud.com.
2. Navigasi ke menu: SoftWare Repository for Container (SWR)> Organization > Create Organization.

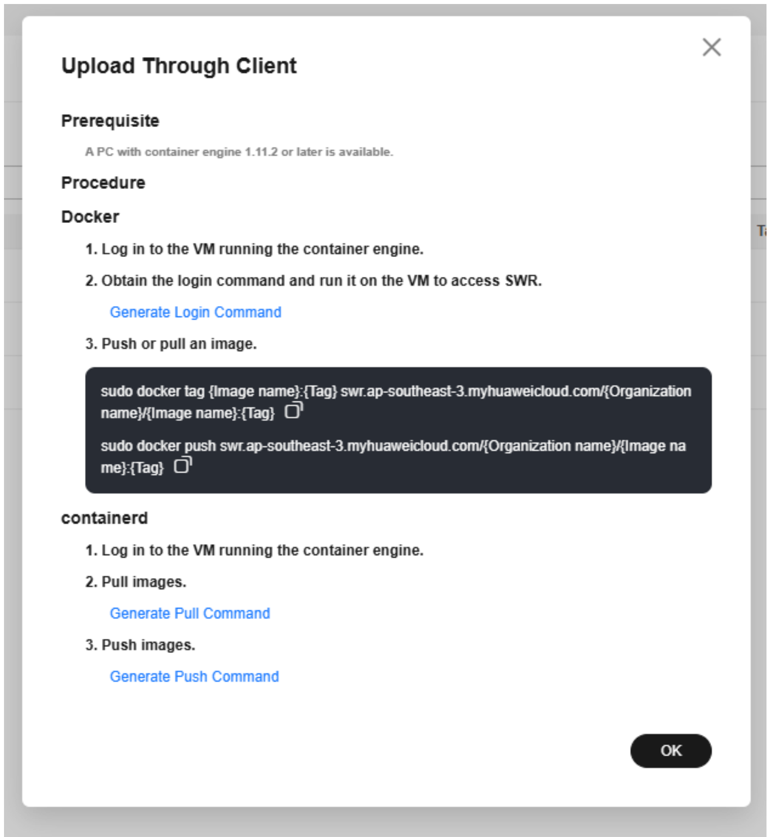


docker login swr.ap-southeast-3.myhuaweicloud.com

- Gantilah <region> dengan region tempat SWR kamu berada, misalnya: cn-north-4, ap-southeast-3, dll.
- Username dan password-nya bisa ambil dari **akses SWR kamu di Huawei Cloud** (biasanya sama dengan AK/SK jika menggunakan mode IAM).
- Bisa menggunakan Temp Login (Klik Generate Login Command)



Langkah 2: Tag Image Lokal ke SWR Format



Misalnya image lokal Anda bernama websocket-server:latest

docker tag websocket-server:latest swr.ap-southeast-3.myhuaweicloud.com/<namespace>/<repository>:latest

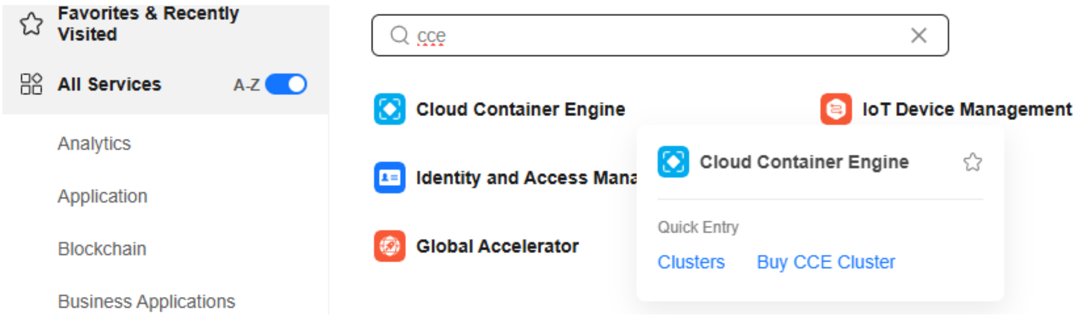
Contoh:

docker tag websocket-server:latest swr.ap-southeast-3.myhuaweicloud.com/aryadev/websocket-server:latest

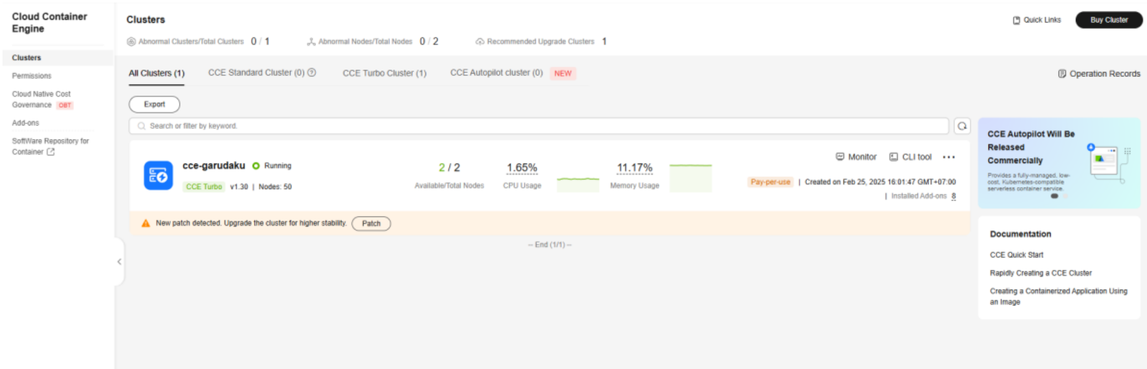
Langkah 3: Push Image ke SWR

docker push swr.ap-southeast-3.myhuaweicloud.com/aryadev/websocket-server:latest

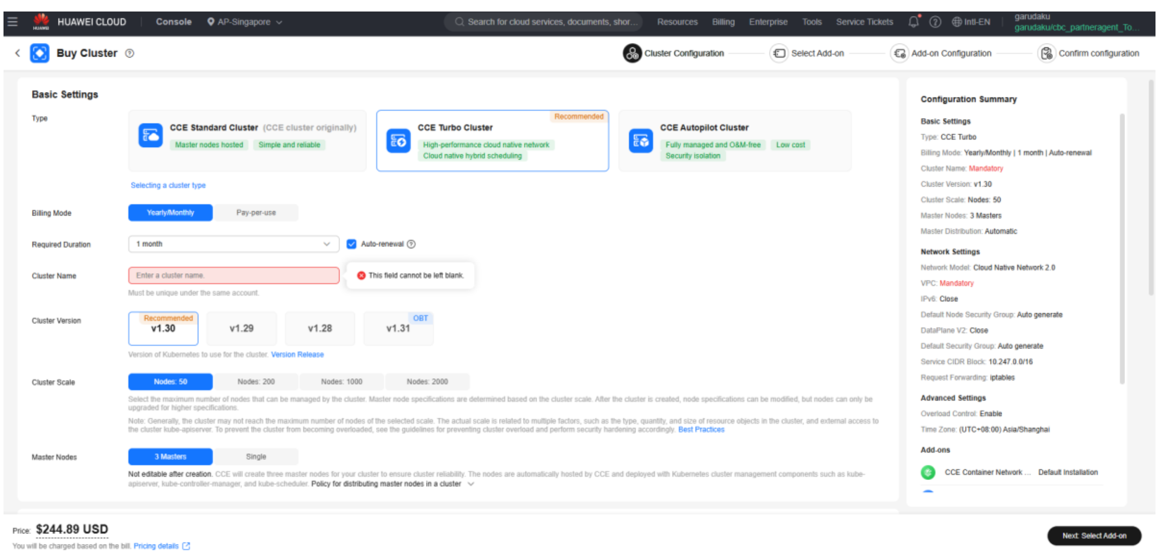
3. Deploy Image ke CCE (Cloud Container Engine)



Langkah 1: Buat Cluster CCE (Jika belum ada)

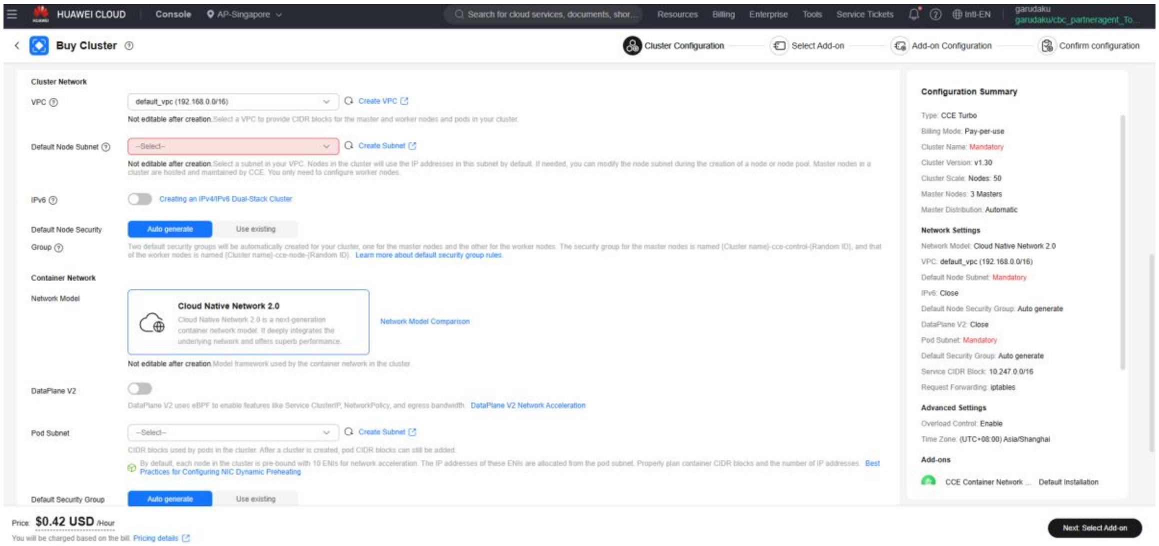


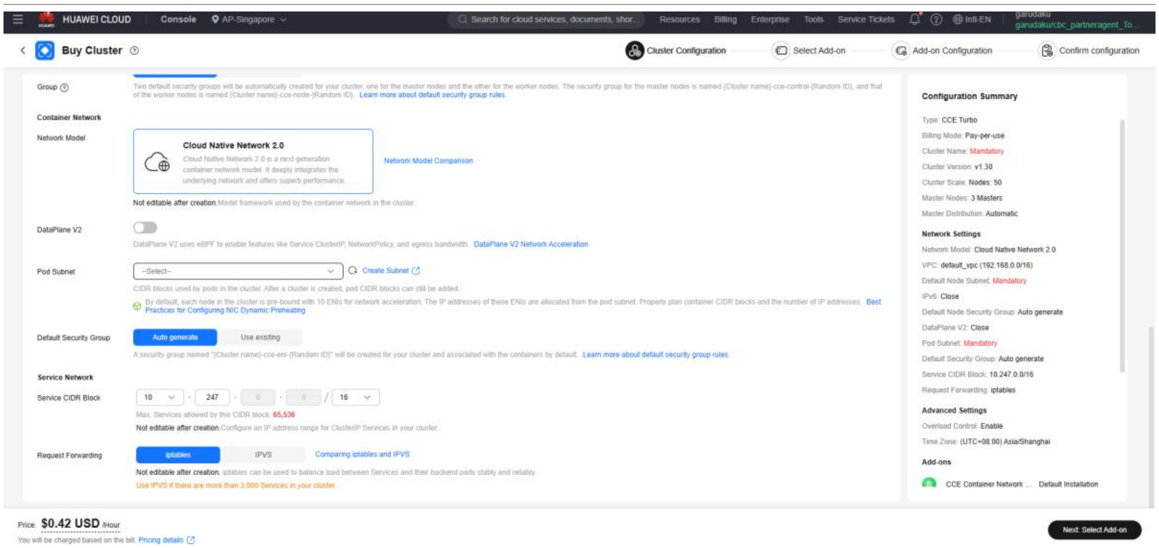
- Masuk ke CCE > Clusters > Buy Cluster



Basic Setting

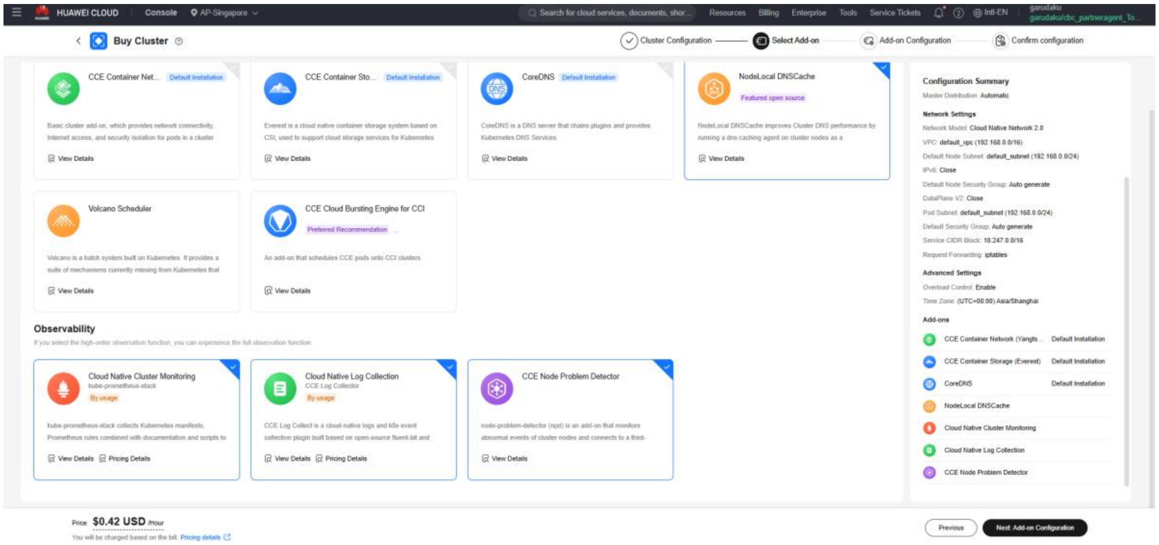
- Pilih region: ap-southeast-3 (Singapore).
- Pilih jenis cluster (misal: Standard , Turbo, AutoPilot Cluster).
- Pilih Billing Mode (misal : Yearly/Montly, PPU/PayPerUse).
- Masukan Nama Cluster.
- Pilih Cluster Scale dan Master Node.





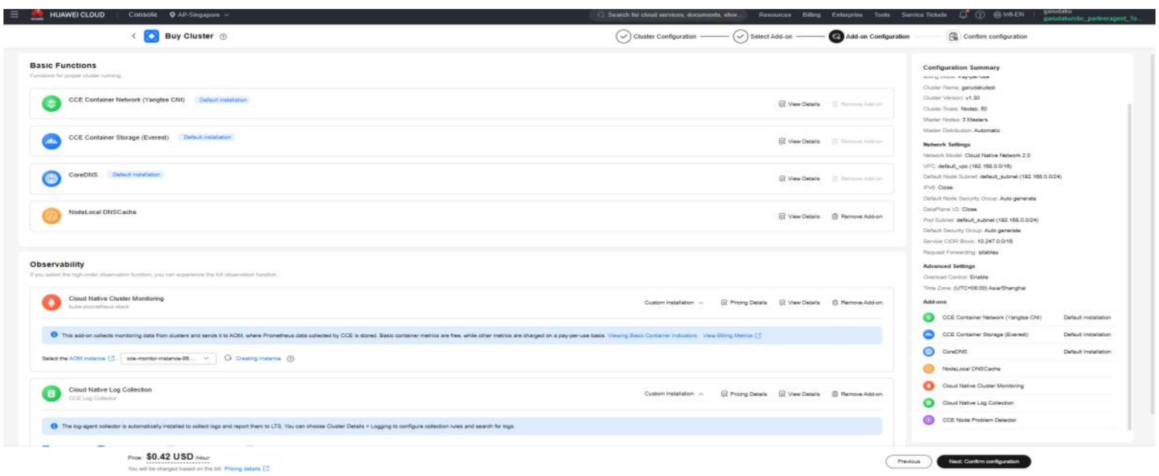
Network Setting

- Pilih VPC yang tersedia (Jika Ada), Kalau tidak ada buat terlebih dahulu untuk VPCnya.
- Pilih Node Subnet yang tersedia (Jika Ada).
- Default Node Security Pilih Auto Generate.
- Pilih Pod Subnet.
- Tentukan Service CIDR Block .
- Next .

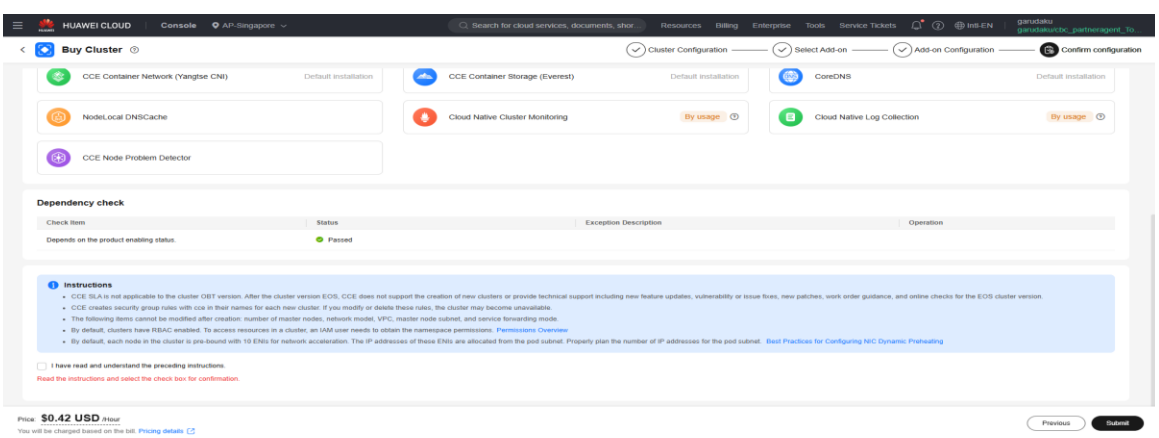


Select Add-on

- Pilih Add-on yang dibutuhkan, Best Practicenya ***Default.***
- Next.



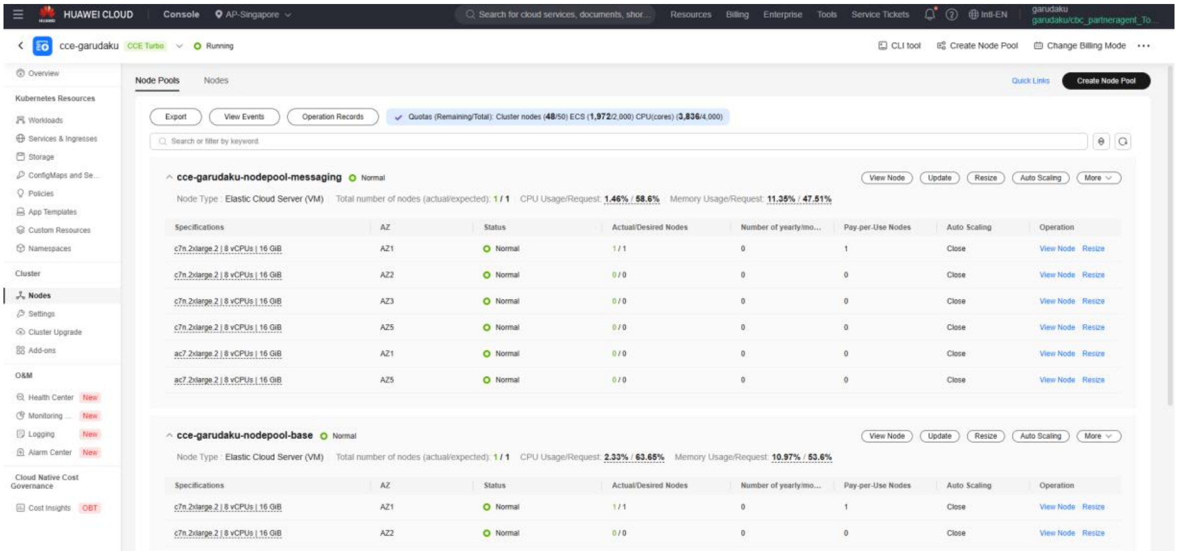
- Next.

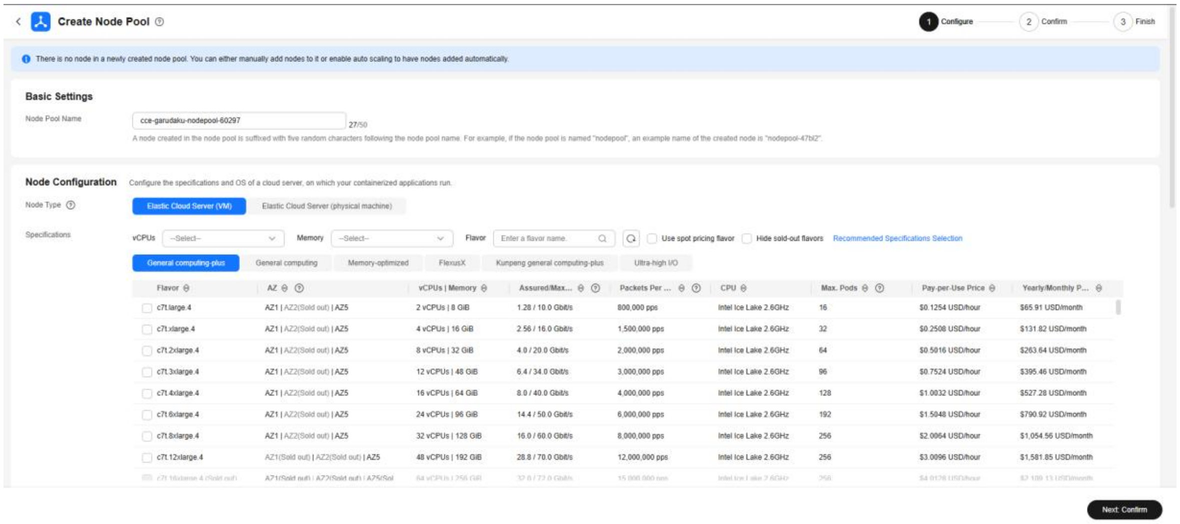


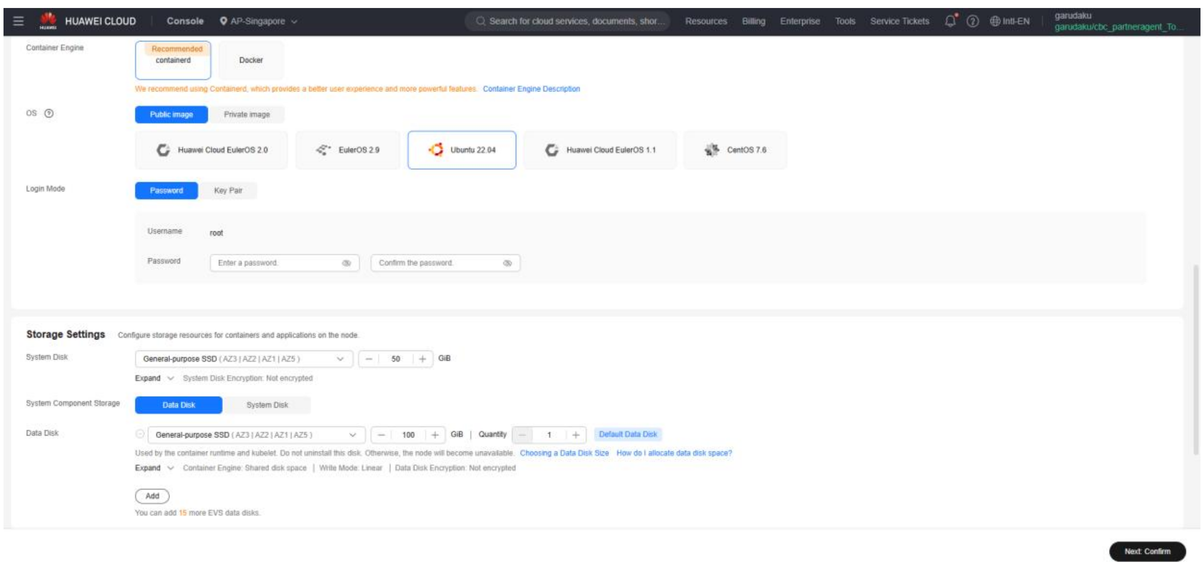
- Submit.

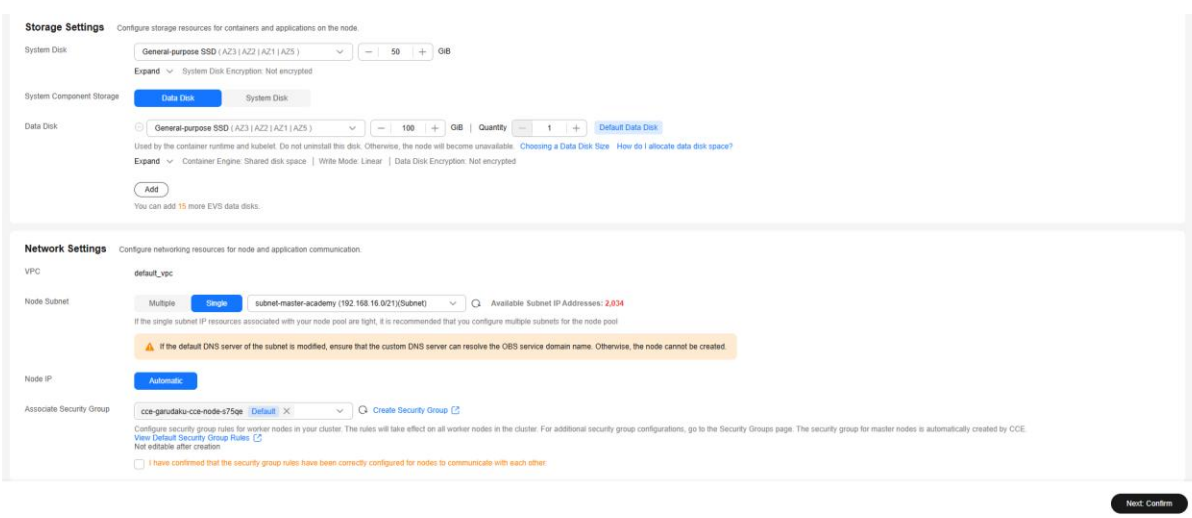
Langkah 2: Configuration Node Pools dan Nodes

1. Masuk Cluster > Nodes > Create Node Pools



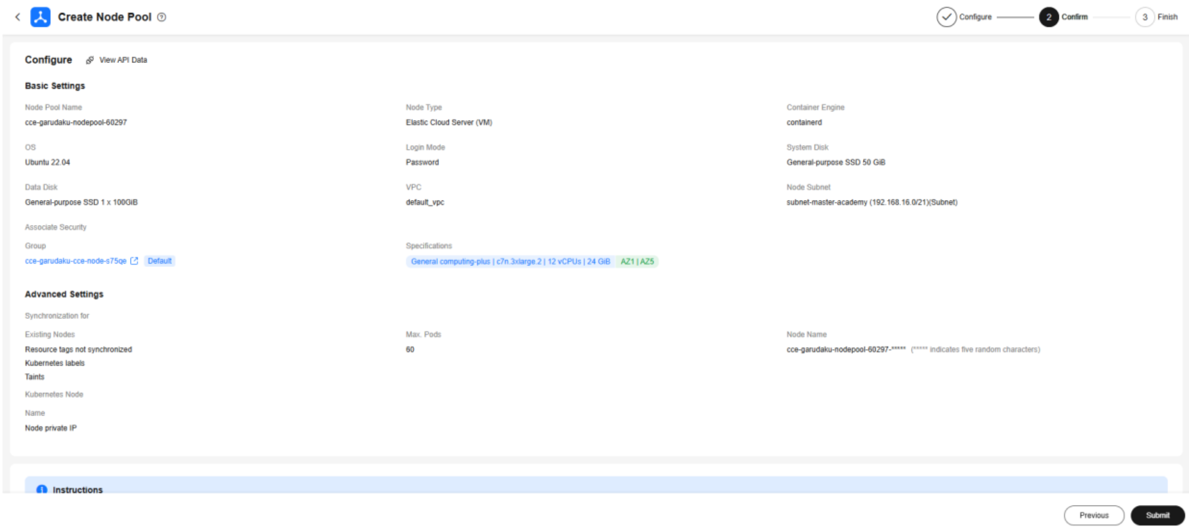






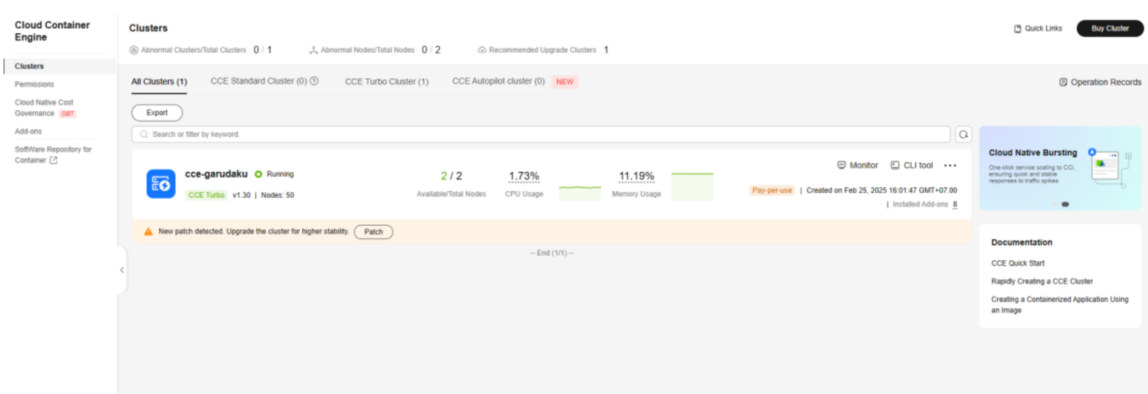
- Tambahkan node pool dan pilih flavor ECS (misal: c6.large.2).
- Tentukan jumlah node yang akan dibuat.

- Pilih image ECS (Ubuntu/CentOS) dan SSH key.
- Tentukan disk system dan data disk sesuai kebutuhan.

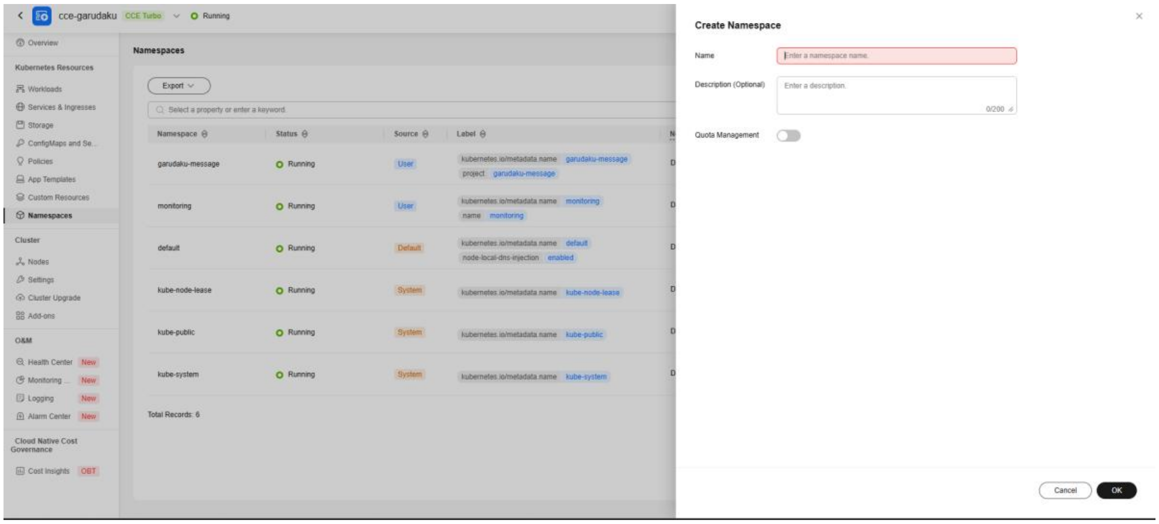


- Klik “Submit” untuk memulai pembuatan cluster.
- Tunggu beberapa menit hingga cluster aktif.

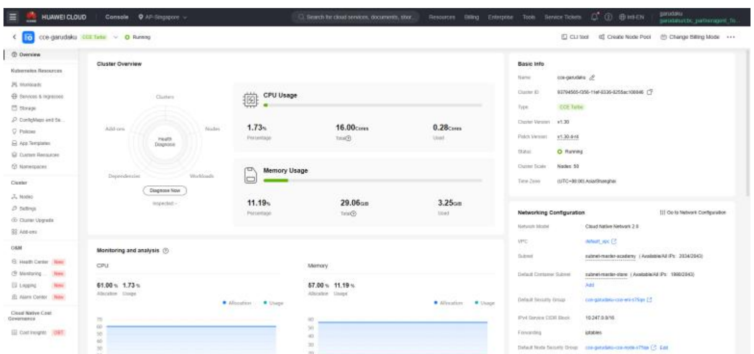
Langkah 3: Buat Deployment di CCE

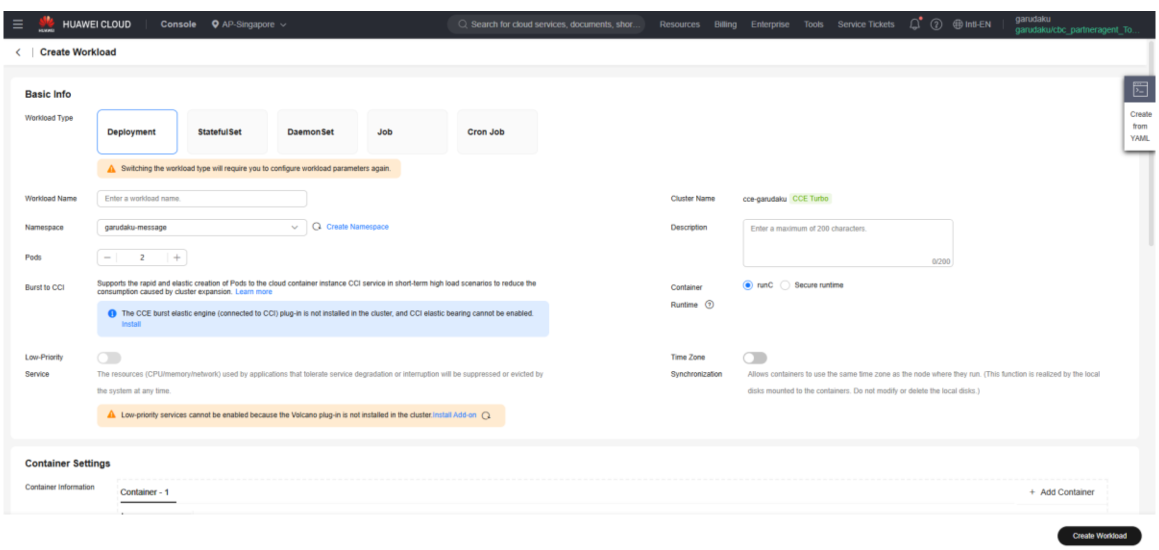


- Masuk ke Cluster > Namescapes > Create Namespace



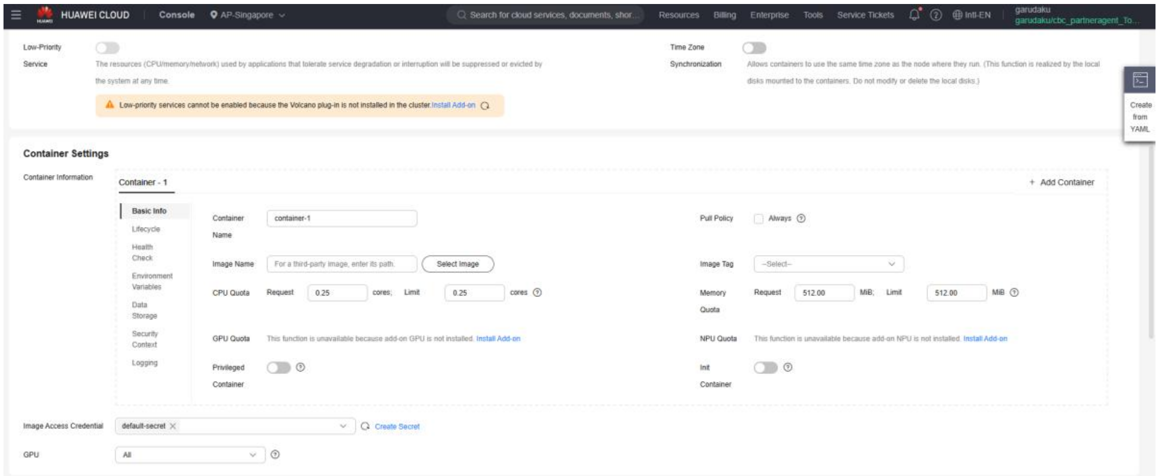
- Masuk ke Cluster > Workloads > Deployments > Create Deployment



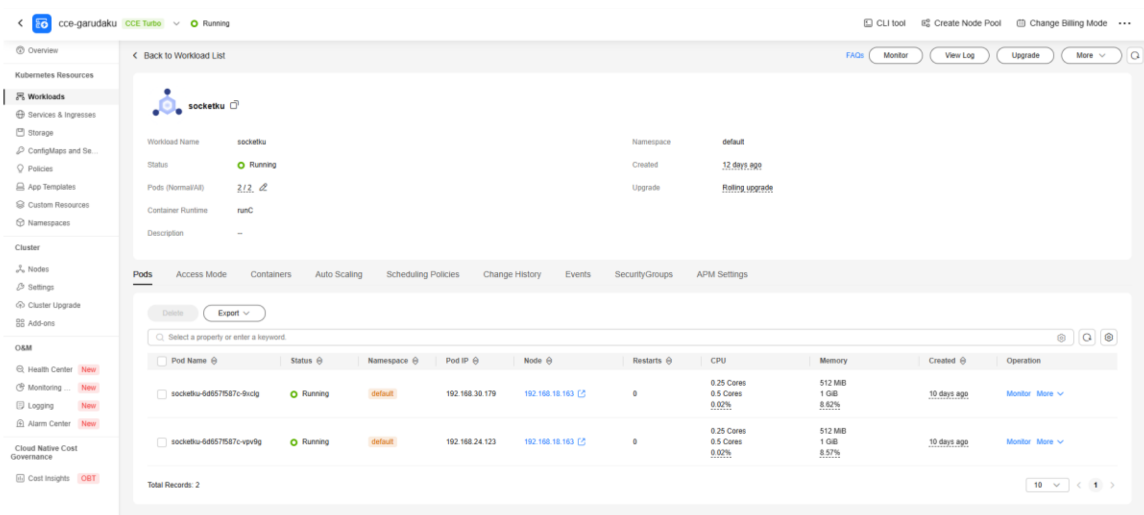


- Isi detail Basic Setting:

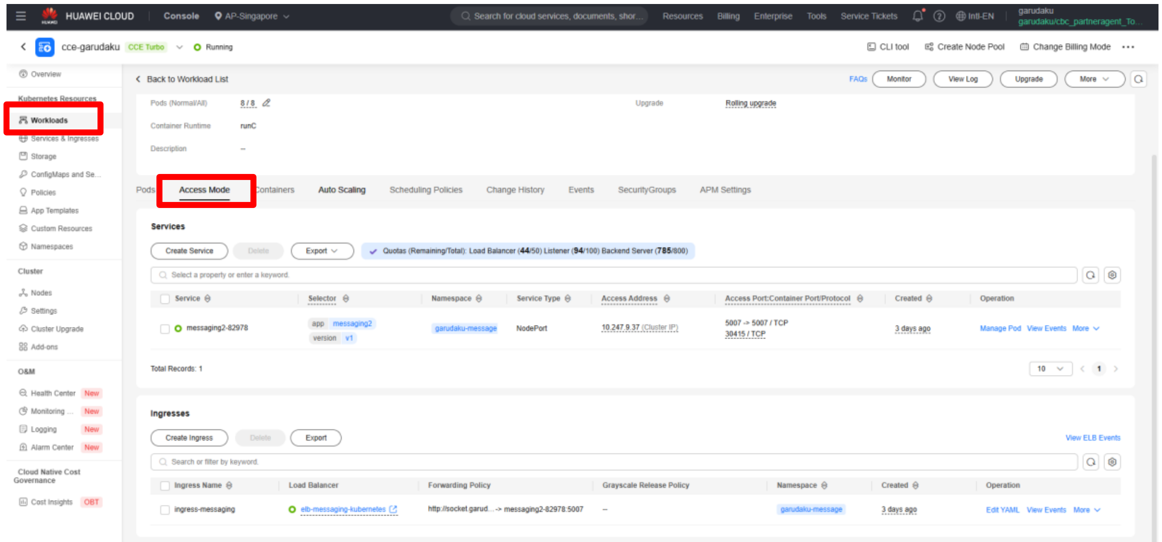
  - Workload Type : Deployment.
  - Workload Name: websocket-garudaku.
  - Pods: misal 2.



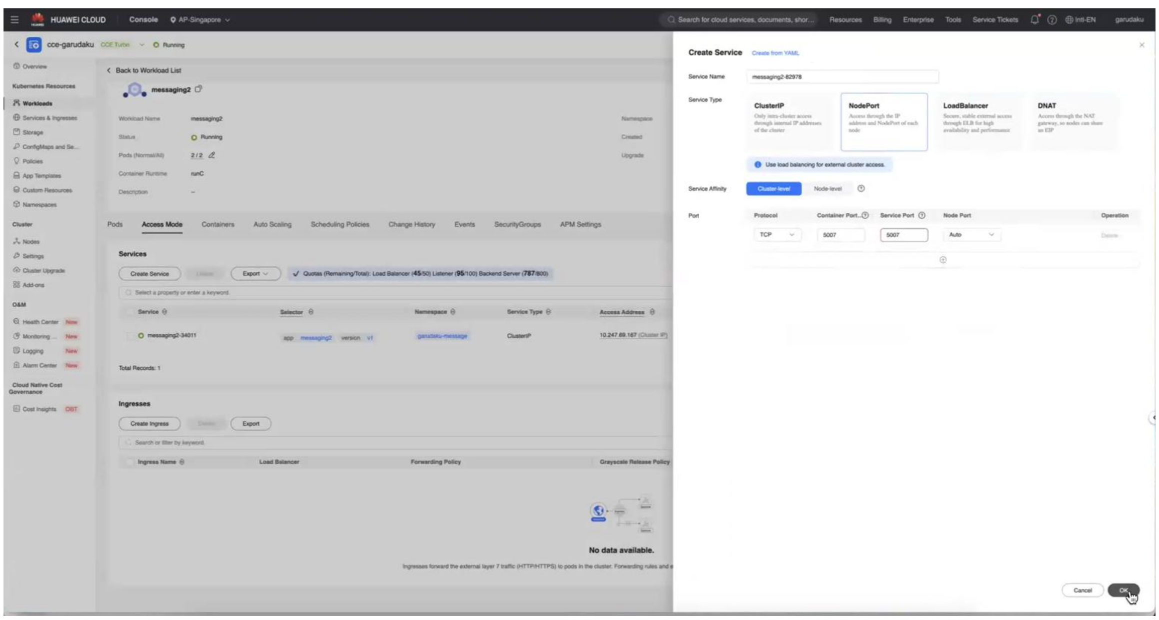
- Container Settings:
  - Container Name : Isi sesuai kebutuhan.
  - Image Name > Klik Select Image > Pilih Image yang sebelumnya di Push ke SWR.
  - Image Tag > Pilih Image Tag yang sebelumnya di Push ke SWR.
  - CPU dan Memory disesuaikan dengan kebutuhan.
  - Untuk settingan berikutnya biarkan Default dan sesuaikan dengan kebutuhan Jika ingin menggunakan PV dan PVC (Container Settings > Data Storage > Add Volume).
  - Jika semua sudah disesuaikan dengan kebutuhan, klik create workload.



Langkah 4: Buat Service dan Ingress



- Masuk ke Workloads > Access Mode > Services > Create Service



- Port Mapping:
  - Service Port: 80
  - ContainerPort: 5007

Catatan Tambahan: Jenis Service Type di CCE Huawei Cloud

1. ClusterIP (default):
- Hanya dapat diakses di dalam cluster.
- Cocok untuk komunikasi antar layanan internal (misalnya: database, backend service).

2. NodePort:
- Mengekspos layanan melalui IP Node dan port tertentu (30000–32767).

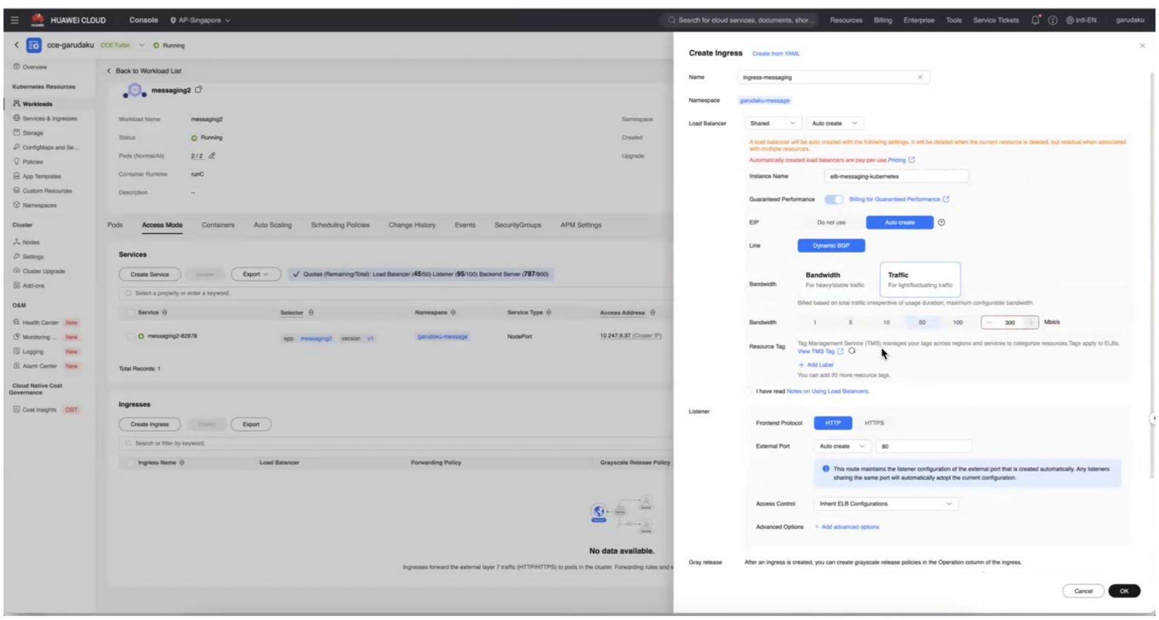
- Dapat diakses dari luar cluster dengan cara: http://<node-ip>:<nodeport>.
- Tidak scalable otomatis, tergantung IP Node ECS.

3. LoadBalancer (Direkomendasikan):
- Membuat Elastic Load Balancer (ELB) secara otomatis.
- Cocok untuk mengakses aplikasi dari luar cluster (publik).
- Akses melalui: http://<public-ip>:<port> atau ws://<public-ip>:80.
- Mendukung akses wss/https dengan konfigurasi tambahan.

5. Ingress (Opsional):
- Digunakan untuk HTTP/HTTPS routing berbasis path atau domain.
- Memerlukan Ingress Controller seperti NGINX.
- Cocok untuk kebutuhan reverse proxy atau routing domain.

Rekomendasi untuk WebSocket: Gunakan Service Type: LoadBalancer. Jika butuh routing domain atau TLS, tambahkan Ingress Controller.

- Masuk ke Workloads > Access Mode > Ingress > Create Ingress



- Nama Ingress : Disesuaikan
- Namespace : Pastikan namespace tidak salah
- Tipe: LoadBalancer (Share/Dedicated – Auto Create)

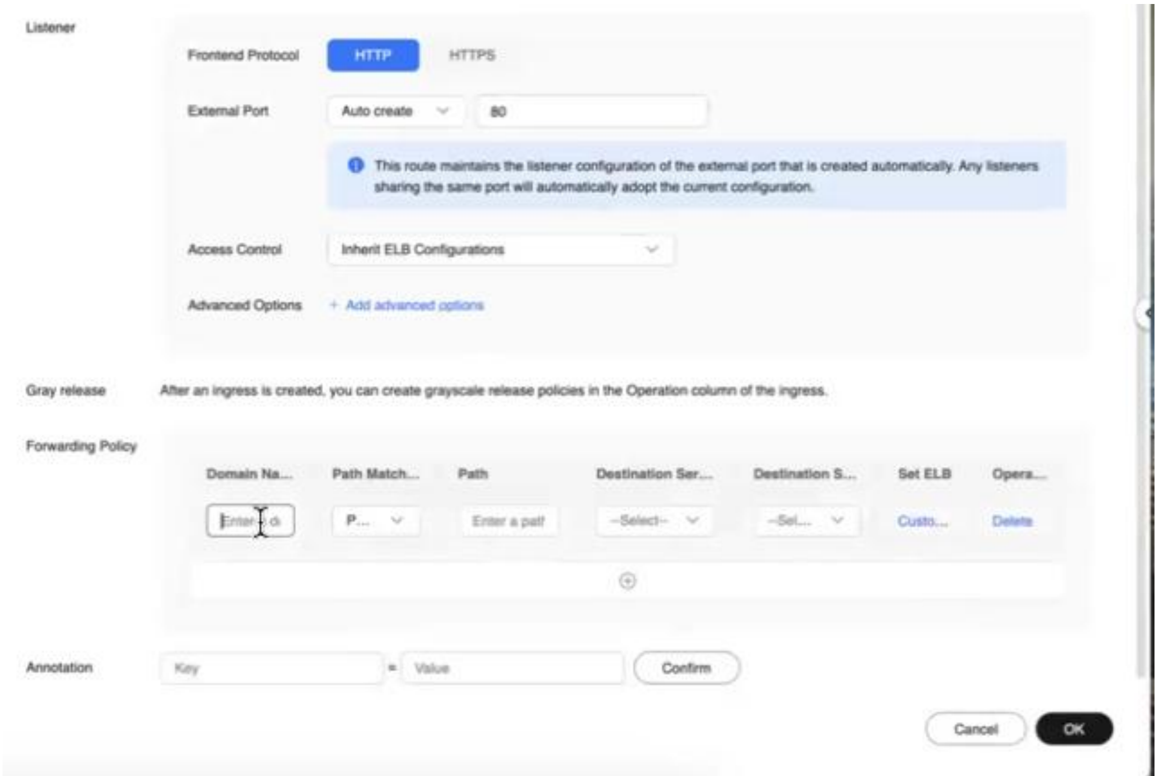
Jika Pilih Shared
- Pilih deployment websocket-deployment
- Instance Name : Disesuaikan kebutuhan.
- Frontend Subnet : Disesuaikan kebutuhan.

- Backend Subnet : Disesuaikan kebutuhan.
- EIP : Auto create

Jika Pilih ELB Shared

Instance name : Disesuaikan dengan kebutuhan

Eip : Auto Create -> Line : Dynamic BGP -> Bandwidth : Traffic 300Mb ->



Listener ELB :

- Frontend Protocol : HTTP/HTTPS (Untuk HTTPS requirmentnya harus menggunakan SSL)
- External Port : Auto Create = 80
- Access Control : Inherit ELB Configuration

Fording policy

Domain Name : Disesuaikan dengan kebutuhan

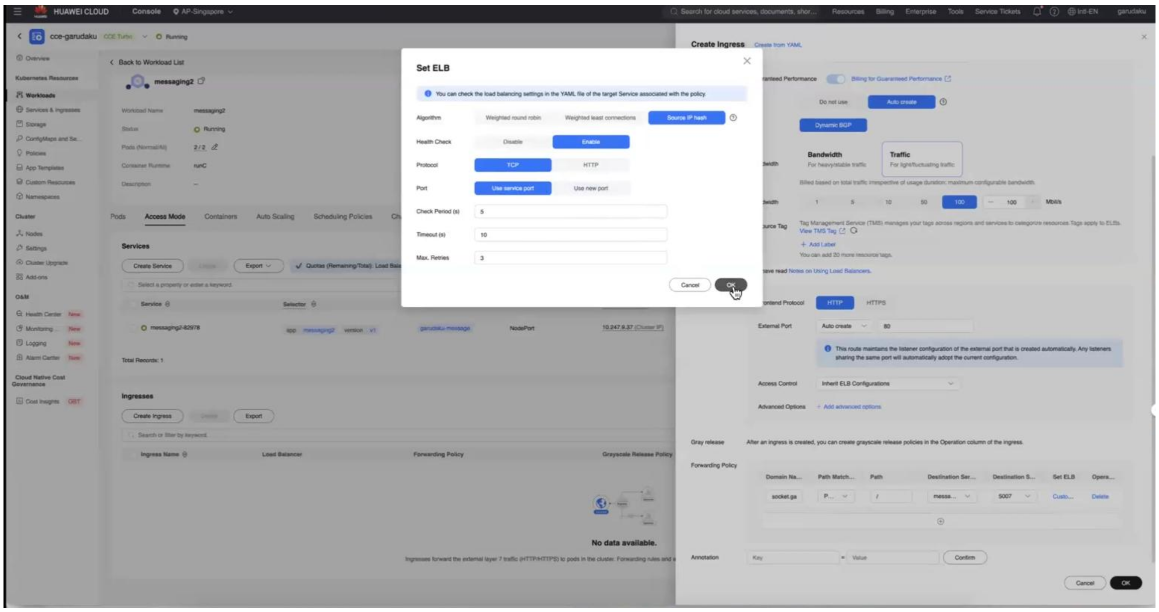
Path Match : Prefix Match

Path : /

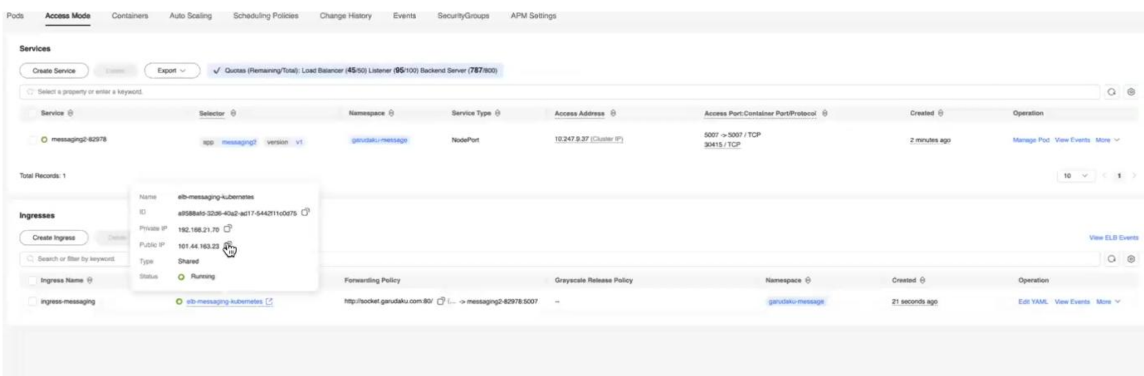
Destination Service : Disesuaikan dengan Service yang sebelumnya telah dibuat

Port : 5007

Set ELB : untuk Web Socket settingannya seperti gambar dibawah ini.



Klik OK. Setelah Service berhasil dibuat, akan muncul public IP address untuk akses WebSocket Anda.



4. Pengujian

- Jika menggunakan domain, maka perlu daftar IP public ELB ke DNS.

- Akses melalui: ws://<public-ip>:80 atau wss:// jika menggunakan HTTPS
- Pastikan aplikasi WebSocket sudah berjalan normal.